

Optimal Mixed-Knowledge Scheduling with SRPT-LAS

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1 Introduction

A key component of scheduling models in queueing is the job size information available to the scheduler. Most models assume sizes are either completely known, unknown, or follow known distributions, giving rise to policies such as First-Come First-Served (FCFS), Least Attained Service (LAS) and the Gittins index policy, respectively [1].

However, practical systems often have a mixture of accurate and inaccurate predictions. Moreover, precise distributional information is rarely available. The ideal scheduling policy exhibits good performance despite these challenges.

A recent paper has addressed scheduling with estimates with unknown distributions in the M/G/1, but that work focused on estimates which were always within a fixed ratio of the true size, rather than a mixture of accurate predictions and inaccurate predictions [7]. Stronger results are known in the batch setting with no arrivals, but not in the M/G/1 [3].

However, important settings remain under studied, including models where some job sizes are exactly known, while some only have inaccurate estimates, as well as models where some job sizes are exactly known, and some are completely unknown. We focus on the model where job sizes are either known or unknown, which we call the “mixed-knowledge model”.

Our goal is to minimize mean response time in the mixed-knowledge M/G/1. To do so, we create the SRPT-LAS policy, which combines the SRPT policy, which is proven to minimize mean response time for known job sizes [5], and the Least Attained Service (LAS) policy, which is proven to minimize mean response time for unknown job sizes under size distributions with decreasing hazard rate [4].

We characterize the mean response time of our SRPT-LAS policy with a tagged job approach. Using this result, we show in Theorem 3.1 that the mean response time under SRPT-LAS converges to the mean response time of SRPT as the proportion of known-size jobs tends to 1.

Only specific scheduling policies have this optimality result. For instance, under SRPT-FCFS, mean response time is infinite under job size distributions with infinite variance, for any fraction of unknown-size jobs, as with FCFS [2].

2 Model of SRPT-LAS Scheduling

We consider an M/G/1 queueing system: jobs arrive according to a Poisson process with rate λ . Each job is one of two types, with different availability of size information:

Known-Size Jobs (Type k): With probability p_k i.i.d., the job’s exact size is known to the scheduler on ar-

rival. The size of these jobs is sampled i.i.d. from a distribution with random variable S_k .

Unknown-Size Jobs (Type u): Otherwise, a job’s size is entirely unknown upon arrival. The size of these jobs is sampled i.i.d. from a distribution with random variable S_u .

The size of a general arrival to the system is denoted by the random variable S . We allow the scheduling policy to serve any job at any time, with preempt-resume service.

For our main result, Theorem 3.1, we consider a sequence of systems where λ and S is fixed, p_k converges to 1, and S_u and S_k change arbitrarily.

To manage this mixed-knowledge workload, we propose a hybrid scheduling policy named SRPT-LAS. The policy always serves the job in the system with the lowest rank, defined as follows: for known-size jobs, the rank is the remaining processing time, as in SRPT, while for unknown-size jobs, the rank is the job’s age, as in LAS. A job’s rank is determined by its age, size, and type, so SRPT-LAS is a SOAP policy [8].

3 Results

THEOREM 3.1. *In the mixed-knowledge M/G/1 with fixed arrival rate λ and overall size distribution S , and arbitrary S_k and S_u distributions, in the limit as the fraction p_k of known-size jobs trends toward 1, the mean response time of SRPT-LAS converges to optimal:*

$$\lim_{p_k \rightarrow 1} (E[T^{SRPT-LAS}] - E[T^{SRPT}]) = 0$$

To prove this result, we characterize the difference in the mean response time of a tagged job with size x in our SRPT-LAS system, and compare against the known result for the SRPT system [6].

We split this difference into three components: waiting time, residence time of known-size jobs, and residence time of unknown-size jobs, and prove each difference vanishes in the $p_k \rightarrow 1$ limit, when integrated across all x .

In future work, we hope to study models with a small fraction of severely mispredicted jobs, replacing the unknown-size jobs in this work, and evaluate the performance of the SRPT-LAS scheduling policy in a multi-server setting.

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